

SAMPLE PATIENT & HEALTHCARE FACILITY DATA

Smart Community Healthcare Access Platform

Purpose: Sample data prepared for understanding the project problem and for initial DSA work.

Note: All names, records, distances and values are fictional sample data created for the PBL project. They do not represent real patients or real healthcare facilities.

Project focus: Organising patient/facility records using trees and representing the local healthcare network using graphs.

1. Sample Patient Records

Six sample patient records were prepared to understand how patient information can be stored and searched using tree structures such as BST and AVL. Patient ID can be used as the key.

Patient ID	Name	Age	Gender	Blood Group	Condition	Priority
P101	Aarav Sharma	45	M	B+	Fever	Normal
P104	Priya Verma	32	F	O+	Asthma	Normal
P107	Rohan Kumar	61	M	A+	Chest Pain	Emergency
P110	Ananya Singh	28	F	AB+	Migraine	Normal
P113	Vivek Gupta	54	M	B-	Diabetes	Priority
P116	Neha Yadav	39	F	O-	Infection	Normal

DSA use: Patient IDs can be inserted into a BST/AVL tree. In-order traversal can later display records in sorted Patient ID order.

2. Sample Healthcare Facility Records

Five sample facilities were selected: two hospitals, one clinic and two pharmacies.

Facility ID	Facility Name	Type	Locality	Doctors	Beds	24x7
F01	City Care Hospital	Hospital	Model Town	18	80	Yes
F02	Green Valley Hospital	Hospital	Civil Lines	12	50	Yes
F03	Sunrise Clinic	Clinic	Shastri Nagar	5	10	No
F04	HealthPlus Pharmacy	Pharmacy	Model Town	-	-	Yes
F05	Wellness Pharmacy	Pharmacy	Civil Lines	-	-	Yes

3. Sample Healthcare Facility Network

Each facility is treated as a graph node and each road connection as an edge. The following distances are fictional values used only for DSA demonstration.

From	To	Distance (km)
F01 – City Care Hospital	F03 – Sunrise Clinic	2.0
F01 – City Care Hospital	F04 – HealthPlus Pharmacy	1.2
F03 – Sunrise Clinic	F04 – HealthPlus Pharmacy	1.0
F02 – Green Valley Hospital	F03 – Sunrise Clinic	2.5
F02 – Green Valley Hospital	F05 – Wellness Pharmacy	1.0
F03 – Sunrise Clinic	F05 – Wellness Pharmacy	1.8

4. Adjacency Matrix

The matrix represents the same five facilities. 0 means there is no direct connection in this sample network.

	F01	F02	F03	F04	F05
F01	0	0	2.0	1.2	0
F02	0	0	2.5	0	1.0
F03	2.0	2.5	0	1.0	1.8
F04	1.2	0	1.0	0	0
F05	0	1.0	1.8	0	0

5. Adjacency List

The same network can also be stored as an adjacency list. It stores the connections that actually exist.

Facility	Connected Facilities
F01 – City Care Hospital	F03 (2.0 km), F04 (1.2 km)
F02 – Green Valley Hospital	F03 (2.5 km), F05 (1.0 km)
F03 – Sunrise Clinic	F01 (2.0 km), F02 (2.5 km), F04 (1.0 km), F05 (1.8 km)
F04 – HealthPlus Pharmacy	F01 (1.2 km), F03 (1.0 km)
F05 – Wellness Pharmacy	F02 (1.0 km), F03 (1.8 km)

6. How This Data Supports the PBL Work

- Patient records provide sample data for BST/AVL record organisation.
- Patient IDs can be used for insertion, searching and traversal examples.
- Facility records show hospitals, a clinic and pharmacies in the sample community.
- The five facilities provide graph nodes for adjacency matrix/list representation.
- The small dataset is suitable for Month 1 manual understanding and can be expanded later.
- Graph traversal and shortest-path features are not claimed as completed work here; they can be added in a later stage.

Dataset status: Initial/sample data for Month 1 understanding and DSA representation. This is not a real hospital database.